SLIIT ENGINEERING • INDEPENDENT REVISION EDITION

ME 1060 DYNAMICS

Expanded learning edition • Weeks 1–6 + Tutorial 6 angular extension

Understand every step: definitions, physical intuition, textbook equations, worked tutorial reasoning, method selection, diagrams, and error checks.

Prepared for: independent revision and exam-style answer construction
Companion: two-sided A4 legal reference sheet

Based on all six uploaded weekly lectures and all six supplied tutorial-solution PDFs. Tutorial 6 adds angular momentum even though the Week 6 lecture focuses on linear impulse and momentum.

Start here: the entire module on one map

Dynamics consists of **kinematics** (what motion occurs?) and **kinetics** (what forces produce it?). The six weeks add several ways of attacking the same physical situation. The hardest part is usually **choosing the right approach**, not remembering an isolated equation.

If the question asks…	Use first	Why
Position, time, speed or trajectory; known motion functions	Kinematics: 1D, x–y, n–t or polar	Connect displacement, velocity, acceleration.
Normal force, tension, friction or instantaneous acceleration	Free-body diagram + ΣF=ma	Forces directly determine acceleration.
Speed after moving a given distance; spring compression	Work–energy / potential energy	Energy directly connects positions without needing time.
Power, motor, efficiency	F·v plus force balance	Power is work done per unit time.
Collision, explosion, large force acting briefly	Impulse–momentum	Time-integrated force changes momentum.
Motion about a point with no external moment	Angular momentum	The component about that point/axis is conserved.

Use your permitted reference sheet to retrieve equations during the exam. Use this guide to learn *when and why* to apply them, along with the complete written-solution structure shown by the tutorials.

How to learn this subject without memorising blindly

This is the companion to the two-sided reference sheet. The sheet tells you *what to write*; this guide tells you *what to think*. The main aim is to get comfortable turning an unfamiliar diagram into the correct equations. A formula without its conditions is dangerous, so each section separates **meaning → choice → mathematical model → solution → physical check**.

> **Four questions to ask before touching your calculator:**
>
> 1. **What is moving?** One particle, multiple connected blocks, a rotating arm, or a system that explodes?
> 2. **What information is given?** A time function, geometric path, force, speed at two positions, or a short force-duration event?
> 3. **Which quantity changes?** Velocity magnitude, direction, energy, linear momentum, or angular momentum?
> 4. **What is actually required?** Instantaneous acceleration (force law), speed over distance (energy), velocity after impact (momentum), or time (kinematics after finding speed)?

Question clue	First thought in your head	Governing approach
“At t = …”, a function of time given	Differentiate or integrate with respect to time.	v = ds/dt, a = dv/dt
“At a location”, curve/path specified	Path geometry determines the useful coordinates.	x–y, n–t, or r–θ
“Force on a rod/track/coupling”	Draw only actual forces; acceleration must agree with motion.	ΣF = ma + constraints
“After moving x metres”, spring and gravity	Do not solve for time if it is not asked.	Work–energy / T + V
“Power required at this instant”	Power needs force and velocity at the same instant.	P = F·v and efficiency
“Very brief explosion/impact”	Separate internal and external impulses.	Σmv before = Σmv after
“Central cable changes the radius”	Force passes through axis: zero torque there.	H about that axis + energy

First learn the meaning of the symbols

THE MOTION CHAIN

$$s \longrightarrow v \longrightarrow a$$

s = signed position along path (m); **v** = velocity (m/s); **a** = acceleration (m/s²); **t** = time (s). Integration travels back from acceleration to velocity to position. A dot on a letter denotes differentiation with respect to time, e.g., $\dot{r}$ = dr/dt.

THREE COMPLEMENTARY PHYSICAL PRINCIPLES

$$\sum \vec{F} = m\vec{a} \qquad T_1 + \sum U_{1\to 2} = T_2$$

$$m\vec{v}_1 + \int_{t_1}^{t_2} \sum \vec{F}\, dt = m\vec{v}_2$$

F = external force (N); **m** = mass (kg); **T** = kinetic energy (J); **U** = work (J); **v** = velocity vector (m/s). These three statements describe the same Newtonian dynamics, but each removes different unwanted unknowns: forces and acceleration, time-free energy, or short-event force impulses.

The sign convention is part of the solution, not decoration

A negative answer does not automatically mean you made a mistake. It means the actual direction is opposite to the positive axis you assumed. For straight-line equations choose **one positive direction**; for components pick x right and y up unless another choice simplifies the problem. For polar coordinates, +r is outward from O and +θ is counterclockwise by convention.

Important distinction: zero speed at a turning point does **not** mean zero acceleration. Constant speed does **not** mean zero acceleration on a curved path. Zero displacement in one direction does not imply zero force in every direction.

Week 1 — Foundations and motion in a straight line

Source focus: Week 1 pp. 6–8, 16–32; Tutorial 1 questions 1–4.

Historical framing from the introductory slides: Galileo developed systematic observations of free fall, inclines and pendulums; Huygens improved time measurement with the pendulum clock; Newton formulated the laws of motion and universal gravitation. The later development of mechanics included Euler, D'Alembert, Lagrange, Laplace, Poinsot, Coriolis and Einstein. Dynamics supports vehicles, robotics, turbines, hoists, machine tools, moving structures, shocks and spacecraft. These points are context rather than substitute equations.

1.1 Fundamental language

Mechanics studies bodies subjected to forces. **Statics** covers bodies in equilibrium; **dynamics** covers accelerated motion. **Kinematics** concerns motion alone; **kinetics** explains the effect of force. **Space** locates an object, **time** orders events, **mass** measures inertia, and **force** is a vector interaction. A **particle** is a useful model where size and rotation can be neglected, while a rigid body does not deform appreciably.

The equations of Newtonian mechanics in these slides are applied in an **inertial frame**: axes fixed in space or translating at constant velocity. A car accelerating forward is not an inertial frame for direct $\Sigma F=ma$ without extra treatment.

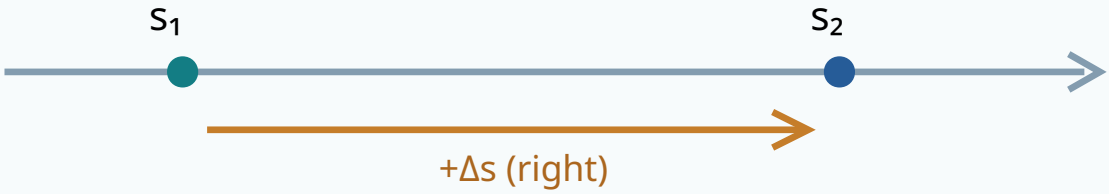
1.2 Signed displacement, speed, velocity and acceleration

> $\Delta s=s_2-s_1$; $v=ds/dt$; $a=dv/dt=d^2s/dt^2=v(dv/ds)$.

Displacement is final position minus initial position and can be negative. Distance counts the entire path length and cannot be negative. Velocity is the **signed rate of displacement**, while speed is its magnitude. Acceleration is the rate of change of velocity, not simply "how fast something moves."

For motion initially forward, negative acceleration may slow it down. Once the velocity becomes negative, that same negative acceleration makes it move backward faster. Therefore always distinguish **sign of a** from **sign of v**.

> Average velocity = $(s_2-s_1)/(t_2-t_1)$; average speed = total distance/(t_2-t_1).



A chosen positive axis gives meaning to signs. An answer of negative velocity describes direction, not an error.

1.3 Identify the mathematical input

What you are given	What to do
v(t)	Integrate v to get s(t); differentiate v to get a(t).
a(t)	Integrate a for v(t), then integrate v for s(t).
a(s)	Use $a=v\,dv/ds$, so $v_2^2-v_1^2=2\int a(s)ds$.
v(s) and time required	$dt=ds/v(s)$, integrate with signed consistency.
Constant acceleration	Use the shortcut kinematic equations below.

For constant a: $v=v_0+at$; $s=s_0+v_0t+\frac{1}{2}at^2$; $v^2=v_0^2+2a(s-s_0)$; $s-s_0=\frac{1}{2}(v_0+v)t$.

Method you should write in the exam

1. Choose one coordinate positive and write the given initial position and velocity with signs.
2. Decide whether acceleration really is constant. If not, use differentiation/integration.
3. Select a relation that directly includes the target while excluding extra unknowns.
4. Solve algebraically and explain the meaning of any negative result.

Exam remark. Those four constant-acceleration formulas are **not valid** for a spring-driven collar whose force varies with location. In that case, use Newton at one instant or energy over a displacement.

Worked model 1 — Tutorial 1, Q1: car with v(t)

Given $v=3t^2+2t$ ft/s, $s(0)=0$. Find position and acceleration at $t=3$ s.

Reasoning: velocity is a function of time; integrate once for position. Acceleration is its derivative.

$s(t)=\int(3t^2+2t)dt=t^3+t^2+C$; $s(0)=0 \Rightarrow C=0$.
$s(3)=27+9=$**36 ft**.
$a(t)=dv/dt=6t+2 \Rightarrow a(3)=$**20 ft/s²**.

Why this structure? Start with the governing derivative, show the initial condition, then evaluate. This matches the tutorial answer order.

Worked model 2 — Tutorial 1, Q2: rocket after engine failure

Rocket speed 75 m/s upward at height 40 m; acceleration is −9.81 m/s² until impact. Find maximum height and impact speed.

Take +y upward. At the highest point vertical velocity is zero. Since time is not needed, use $v^2=v_0^2+2a\Delta y$.

$0=75^2+2(-9.81)(y_{max}-40) \Rightarrow \mathbf{y_{max}=327\ m}$.
From top to ground: $v^2=0+2(-9.81)(0-327)$.
$v=\mathbf{-80.1\ m/s} \Rightarrow$ speed **80.1 m/s downward**.

Why the minus sign? Velocity is directed downward along the chosen upward-positive axis. The speed itself remains positive.

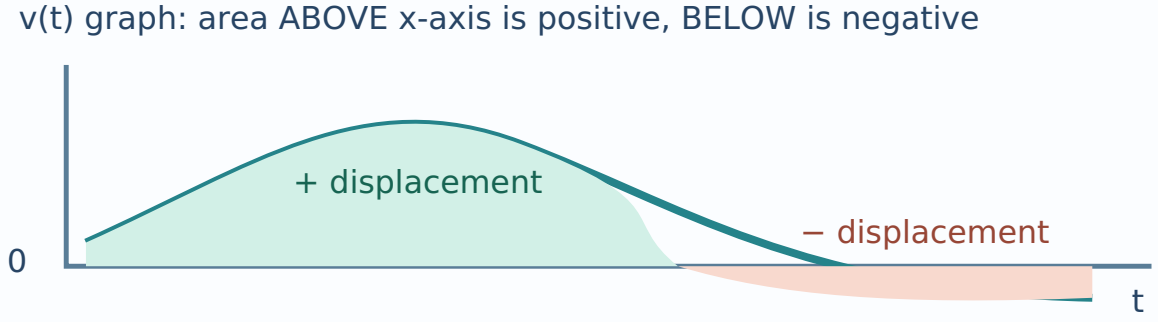
1.4 Curvilinear motion starts with vector derivatives

$\vec{r}=\vec{r}(t)$; $\Delta\vec{r}=\vec{r}_2-\vec{r}_1$; $\vec{v}=d\vec{r}/dt$ (tangent to path); $\vec{a}=d\vec{v}/dt=d^2\vec{r}/dt^2$.

In a curve, velocity is tangent to the path but acceleration need not be. Acceleration can change the speed, the direction, or both. This motivates Cartesian, tangent-normal and polar descriptions in Week 2.

1.5 Why you differentiate, why you integrate

Think of velocity as the **slope of a position-versus-time graph**. An upward-sloping position curve has positive velocity. Acceleration is the slope of the velocity graph. The **signed area** under a velocity-versus-time graph equals displacement, while the signed area under acceleration-versus-time equals the change in velocity. To obtain total *distance* from a velocity graph, you must treat areas below the axis as positive distances.



The actual drawn curve is schematic. A particle changes direction when v crosses zero; split the integral there to get total distance.

1.6 Worked lecture model: when velocity changes sign

Lecture 1, pp. 33–34 — displacement is not distance

Particle velocity is $v(t) = 3t^2 - 6t$ m/s, and $s(0) = 0$. Find its position, total distance, average velocity and average speed over $0 \le t \le 3.5$ s.

Think: Find position by integrating. But the word *distance* requires a sign check: first solve $v(t) = 0$ and divide the journey into forward/backward segments.

STEP 1 · INTEGRATE AND LOCATE REVERSAL

$$s(t) = \int (3t^2 - 6t)\,dt = t^3 - 3t^2 + C$$

$$s(0) = 0 \Rightarrow C = 0, \qquad v(t) = 3t(t-2) = 0$$

Turning times: t = 0 s (start) and t = 2 s (reversal). At t = 2, s = −4 m. At t = 3.5, s = 6.125 m.

Step 2 · Signed displacement: Δs = 6.125 − 0 = **+6.125 m**.

Step 3 · Total distance: |−4−0| + |6.125−(−4)| = 4 + 10.125 = **14.125 m**.

Step 4 · Time averages: v_avg = 6.125/3.5 = **1.75 m/s**; average speed = 14.125/3.5 = **4.04 m/s**.

Why this matters: blindly integrating v from 0 to 3.5 and calling the result “distance” loses marks. Velocity is signed; distance is not.

1.7 General variable-acceleration toolbox: three distinct cases

WHEN ACCELERATION IS GIVEN AS A FUNCTION OF TIME

$$v_2 - v_1 = \int_{t_1}^{t_2} a(t)\,dt$$

$$s_2 - s_1 = \int_{t_1}^{t_2} v(t)\,dt$$

Use these **two sequential integrations** when the time dependence is known; initial conditions determine constants of integration.

WHEN ACCELERATION IS GIVEN AS A FUNCTION OF POSITION

$$a = v\frac{dv}{ds} \quad \Longrightarrow \quad \frac{v_2^2 - v_1^2}{2} = \int_{s_1}^{s_2} a(s)\,ds$$

The trick comes from the chain rule: dv/dt = (dv/ds)(ds/dt). Useful when time is absent from a question.

WHEN ACCELERATION IS GIVEN AS A FUNCTION OF VELOCITY

$$dt = \frac{dv}{a(v)}, \qquad ds = \frac{v\,dv}{a(v)}$$

These are differential relations: integrate only after expressing a as a function of v. Do not substitute a constant-acceleration formula if a varies.

Small habit: after solving v from v², check the geometry to decide the correct sign. A square root gives a speed; direction still comes from the motion and the chosen axis.

Week 2 — Plane kinematics: rectangular, tangent-normal, polar and relative motion

Source focus: Week 2 pp. 3–17 and 19–29; Tutorials 1–2, including trajectory, robot arm, cardioid, pulley and projectile.

2.1 Cartesian x–y coordinates

> $\vec{r}=x\mathbf{i}+y\mathbf{j}$; $\vec{v}=(dx/dt)\mathbf{i}+(dy/dt)\mathbf{j}$; $\vec{a}=(d^2x/dt^2)\mathbf{i}+(d^2y/dt^2)\mathbf{j}$.

The x and y components are independent **equations**, but they refer to the same instant t. Their vector magnitudes are $\sqrt{(v_x^2+v_y^2)}$ and $\sqrt{(a_x^2+a_y^2)}$. Determine direction with a quadrant-aware inverse tangent.

> If y=f(x): $v_y=(dy/dx)v_x$, $a_y=(d^2y/dx^2)v_x^2+(dy/dx)a_x$.

This is the chain rule, useful when the path equation y(x) is known but y(t) is not given directly. It appears in the weather-balloon tutorial.

Worked model 3 — Tutorial 1, Q3: differentiating a path

x=8t ft, y=x²/10 ft. Find $\vec{v}$ and $\vec{a}$ at t=2 s.

At t=2, x=16 ft, dx/dt=8 ft/s and d²x/dt²=0. Use the chain rule for y.

> $v_x=8$; $v_y=(2x/10)(8)=25.6$ ft/s.
> $|\vec{v}|=\sqrt{(8^2+25.6^2)}=$**26.8 ft/s**, direction **72.6° above +x**.
> $a_x=0$; $a_y=(2/10)(8^2)+(2x/10)(0)=$**12.8 ft/s² upward**.

Why use the chain rule? The vertical coordinate depends on time *through x*, so differentiating y with respect to x alone is not enough.

Worked model 4 — Tutorial 2, Q1: find the event time first

$v_x=50-16t$, $y=100-4t^2$; x(0)=0. Find $\vec{v}$ and $\vec{a}$ when y=0.

> $x=\int(50-16t)dt=50t-8t^2$; $a_x=-16$.
> $v_y=-8t$; $a_y=-8$.
> $100-4t^2=0 \Rightarrow$ physical t=**5 s**.
> $\vec{v}=$**$-30\mathbf{i}-40\mathbf{j}$ m/s**, $|v|=50$ m/s.
> $\vec{a}=$**$-16\mathbf{i}-8\mathbf{j}$ m/s²**, $|a|=17.89$ m/s².

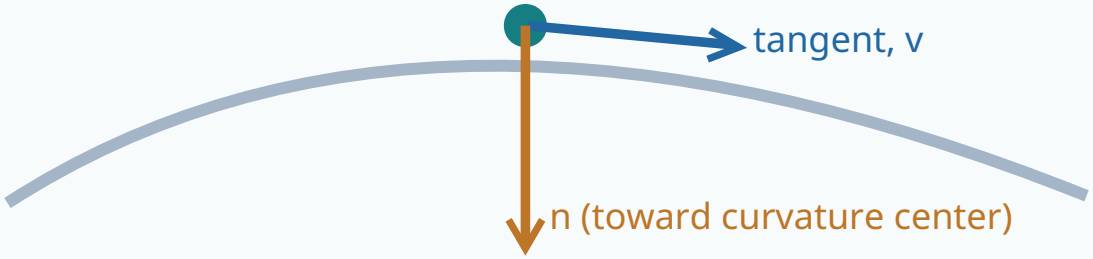
Method choice: solve the event condition y=0 first, then evaluate both components at the same t.

2.2 Tangential–normal coordinates (n–t)

Use n–t when the **path is known**. The unit tangent t points with the instantaneous velocity. The normal n points toward the local center of curvature. Changes in speed are tangential; changes in direction are normal.

> $\vec{v}=v\,\vec{u}_t$; $\vec{a}=(dv/dt)\vec{u}_t+(v^2/\rho)\vec{u}_n$; $\rho=[1+(dy/dx)^2]^{3/2}/|d^2y/dx^2|$.

The normal acceleration $a_n=v^2/\rho$ is never directed away from the center. A vehicle circling at constant speed still has this acceleration. For a circular path, $\rho=R$, $v=R\omega$ and $a_t=R\alpha$ when the direction convention is consistent.



Schematic directions; n must point toward the actual center of curvature at the specific location.

2.3 Polar coordinates (r, θ): rotating arms and sliding collars

Polar axes rotate with the particle. Radial u_r points from O to the particle, while transverse u_θ is 90° counterclockwise from it. Unlike Cartesian basis vectors, these unit vectors change with time.

> $\vec{v}=(dr/dt)\vec{u}_r+r(d\theta/dt)\vec{u}_\theta$; $\vec{a}=[d^2r/dt^2-r(d\theta/dt)^2]\vec{u}_r+[r(d^2\theta/dt^2)+2(dr/dt)(d\theta/dt)]\vec{u}_\theta$.

Interpret each term: $\ddot{r}$ measures radial speed change; $-r\omega^2$ is inward centripetal acceleration; $r\alpha$ is transverse acceleration from angular acceleration; $2\dot{r}\omega$ is the additional Coriolis component associated with radial motion in rotating coordinates.

If the problem gives $r=f(\theta)$, differentiate using the chain rule: $\dot{r}=(dr/d\theta)\omega$ and $\ddot{r}=(d^2r/d\theta^2)\omega^2+(dr/d\theta)\alpha$. Do not make $\dot{r}=0$ simply because ω is constant.

Method you should write in the exam

1. Write the geometric relation $r=r(\theta)$ or $r=r(t)$ before differentiating.
2. Find $\dot{r}$, $\ddot{r}$, $\omega=\dot{\theta}$, $\alpha=\ddot{\theta}$ at the requested instant. Convert rpm to rad/s.
3. Calculate v_r, v_θ, a_r and a_θ.
4. Use vector magnitude and correct quadrant for direction; if forces are asked, continue with FBD and polar $\Sigma F=ma$.

Worked model 5 — Tutorial 2, Q2: telescoping rotating arm

$r=0.500$ m, $\dot{r}=-0.200$ m/s (retracting), $\ddot{r}=0$, $\omega=15$ rev/min$=\pi/2$ rad/s, $\alpha=0$.

$v_r=-0.200$; $v_\theta=0.5(\pi/2)=0.7854 \Rightarrow |v|=$**0.8105 m/s**.
$a_r=0-0.5(\pi/2)^2=-1.2337$; $a_\theta=0+2(-0.2)(\pi/2)=-0.6283$.
$|a|=\sqrt{(1.2337^2+0.6283^2)}=$**1.3844 m/s²**.

Important sign: retraction means $\dot{r}<0$. Tutorial output gives the acceleration magnitude; the negative radial and transverse components explain its actual direction.

Worked model 6 — Tutorial 2, Q4: derivatives of r(t) and θ(t)

A collar has $r=100t^2$ mm and $\theta=t^3$ rad. Find speed and acceleration at t=1 s.

First convert to metres: $r=0.1t^2$ m. Then $\dot{r}=0.2t$, $\ddot{r}=0.2$, $\omega=3t^2$ and $\alpha=6t$.

At t=1: $r=0.1$, $\dot{r}=0.2$, $\omega=3$, $\alpha=6$.
$v_r=0.2$; $v_\theta=0.3 \Rightarrow |v|=$**0.361 m/s**.
$a_r=0.2-0.1(3^2)=-0.7$; $a_\theta=0.1(6)+2(0.2)(3)=1.8$.
$|a|=$**1.93 m/s²**.

Why this structure? The easiest route is to calculate all time derivatives first, then substitute into the memorized polar components.

Extra tutorial walkthrough — Tutorial 2, Q3: cardioid-shaped slot

$r=0.5(1-\cos\theta)$ ft; at θ=180° a ball has speed 4 ft/s and acceleration magnitude 30 ft/s². Find ω and α of the slotted arm.

This is a polar-geometry question, not a constant-radius circle. At $\theta=\pi$: $r=1$ ft, $dr/d\theta=0$, $d^2r/d\theta^2=-0.5$.

$v^2=\dot{r}^2+(r\omega)^2=(0)^2+(1\omega)^2=4^2 \Rightarrow$ **ω=4 rad/s**.
$\ddot{r}=(-0.5)(4^2)+(0)\alpha=-8$ ft/s².
$a_r=-8-1(4^2)=-24$; $a_\theta=1\alpha+0=\alpha$.
$30^2=(-24)^2+\alpha^2 \Rightarrow$ **α=18 rad/s²**.

Why polar? The guide is defined as r(θ) and the unknowns are angular derivatives. Differentiate r(θ) before using the polar acceleration law.

2.4 Projectiles: split into two independent directions

$x=x_0+(u\cos\beta)t$; $y=y_0+(u\sin\beta)t-\frac{1}{2}gt^2$; $v_x=u\cos\beta$; $v_y=u\sin\beta-gt$.

Worked model 7 — Tutorial 2, Q6: horizontal launch

A sack leaves horizontally at 12 m/s from height 6 m.

Choose origin at launch, x to the right and y upward. Its initial vertical speed is zero.

$-6=0+0t-\frac{1}{2}(9.81)t^2 \Rightarrow$ **$t=1.11$ s**.
$R=v_x t=(12)(1.106) \Rightarrow$ **$R=13.3$ m**.

Why vertical motion first? Height is given, so vertical displacement determines flight time; horizontal motion then gives range.

Lecture expansion — Week 2, fixed wheel: radial and tangential acceleration

A fixed wheel has diameter 0.6 m, angular speed 5 rad/s, and angular deceleration 10 rad/s². Determine rim acceleration.

The rim radius is $r=0.3$ m, with $\dot{r}=\ddot{r}=0$. Because the angular acceleration opposes rotation, choose $\alpha=-10$ rad/s² in the chosen positive angular sense.

$a_r=-r\omega^2=-0.3(5^2)=$ **-7.5 m/s² (inward)**.
$a_\theta=r\alpha=0.3(-10)=$ **-3.0 m/s²**.
$|a|=\sqrt{(7.5^2+3.0^2)}=$ **8.08 m/s²**; angle to inward radial $\approx\tan^{-1}(3/7.5)=21.8°$.

Use a signed angular acceleration: deceleration means α opposite ω, not necessarily a negative number before defining a positive sense.

Lecture expansion — Week 2, rotating searchlight on a wall

A searchlight rotates at constant ω=4 rad/s, projecting a point of light onto a vertical wall 100 m away. Find the apparent spot velocity and acceleration at θ=45°.

Instead of using the full polar components, recognize the geometric constraint along the wall: $y=100\tan\theta$. Differentiate with respect to time; $\ddot{\theta}=0$.

$dy/dt=100\sec^2\theta\cdot\dot{\theta}=100(2)(4)=$ **800 m/s**.
$d^2y/dt^2=200\sec^2\theta\tan\theta\cdot\dot{\theta}^2+100\sec^2\theta\cdot\ddot{\theta}$
$=200(2)(1)(4^2)+0=$ **6400 m/s²**.

Why geometry first? The light spot is constrained to move along the wall, so a direct wall-coordinate function is more economical than resolving two polar vector components.

2.5 Relative motion

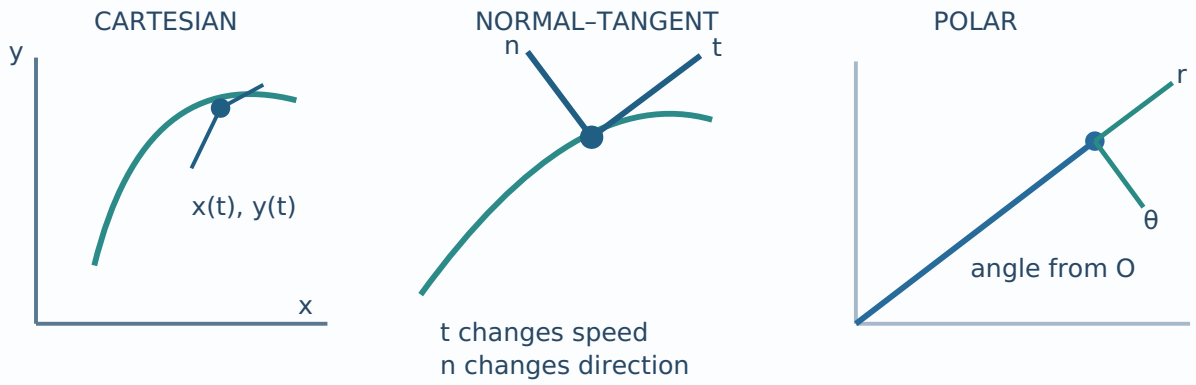
$\vec{r}_A=\vec{r}_B+\vec{r}_{A/B}$; $\vec{v}_A=\vec{v}_B+\vec{v}_{A/B}$; $\vec{a}_A=\vec{a}_B+\vec{a}_{A/B}$.

The subscript A/B means the motion of A as seen from B. These are the translating-axis relations on the lecture slides; for a rotating relative frame extra terms are needed and were not supplied in this week's relative-motion formula set.

Exam remark. The lecture also illustrates geographic Coriolis deflection: to the right of motion in the Northern Hemisphere and to the left in the Southern Hemisphere. For this midterm's polar mechanics questions, apply the mathematical component $2\dot{r}\omega$ and its correct sign.

Tutorial 2 Q5 (belt-driven pulley) provides final answers but no complete calculation in the supplied file: angular acceleration **300 rad/s²**, acceleration of B **37.5 m/s²**, and acceleration of belt point C **22.5 m/s²**. For the method, use no slip to relate belt speed and rim angular speed, split pulley acceleration into $r\alpha$ and $r\omega^2$, and inspect the belt curvature at the specified point. Do not assume a straight belt segment has centripetal acceleration.

2.6 The coordinate-system decision that prevents half the mistakes



Cartesian axes remain fixed; tangent–normal axes follow the path; polar axes rotate with the line from O to the particle. Therefore "radial" and "normal" are not automatically the same direction.

2.7 The full polar derivation, with meaning attached

Polar coordinates are the part many students try to memorise. Instead, notice that when the arm turns, its unit vectors also turn. The derivative of position therefore contains **two** velocity components even when the particle has only one distance r from the origin.

STEP A · ROTATING UNIT VECTORS

$$\frac{d\vec{e}_r}{dt} = \dot{\theta}\vec{e}_\theta, \qquad \frac{d\vec{e}_\theta}{dt} = -\dot{\theta}\vec{e}_r$$

e_r = outward unit vector; $e\theta$ = unit vector 90° counterclockwise from e_r; $\dot{\theta}$ = angular speed (rad/s). The minus sign occurs because $e\theta$ rotates toward $-e_r$.

STEP B · DIFFERENTIATE POSITION ONCE

$$\vec{r} = r\vec{e}_r \quad \Rightarrow \quad \vec{v} = \dot{r}\vec{e}_r + r\dot{\theta}\vec{e}_\theta$$

$\dot{r}$ = sliding speed along arm; $r\dot{\theta}$ = sideways speed from rotation. The two directions are perpendicular.

STEP C · DIFFERENTIATE VELOCITY AGAIN

$$\vec{a} = (\ddot{r} - r\dot{\theta}^2)\vec{e}_r + (r\ddot{\theta} + 2\dot{r}\dot{\theta})\vec{e}_\theta$$

r̈ = radial change of sliding speed; −rθ̇² = centripetal contribution; rθ̈ = angular-acceleration contribution; 2ṙθ̇ = Coriolis-type component in these rotating coordinates.

Check the sign of ṙ. If a telescoping rod is getting shorter, ṙ is negative. The 2ṙθ̇ term may therefore be negative, even while θ̇ is positive. In a circular path with fixed r, ṙ = r̈ = 0, but −rθ̇² remains.

2.8 Differentiating an equation of the path r = f(θ)

The geometry might give a cardioid, a slot, or a line. Since r depends on θ and θ depends on t, you must use the **chain rule twice**:

USE WHENEVER R IS SUPPLIED IN TERMS OF Θ

$$\dot{r} = \frac{dr}{d\theta}\dot{\theta}$$

$$\ddot{r} = \frac{d^2r}{d\theta^2}\dot{\theta}^2 + \frac{dr}{d\theta}\ddot{\theta}$$

dr/dθ tells how radius changes per radian; θ̇ turns that into metres per second. The second expression has two terms: curvature of the r–θ relationship and change of angular speed.

Cardioid-slot tutorial method (Tutorial 2 Q3):

1. Use the path equation r(θ) to calculate r, dr/dθ and d²r/dθ² at the specified angle.
2. Write ṙ = (dr/dθ)θ̇. Construct v² = ṙ² + (rθ̇)² to determine the permissible θ̇.
3. Construct r̈ by the chain rule above; insert r̈ into a_r = r̈ − rθ̇².
4. Write aθ = rθ̈ + 2ṙθ̇ and match with the physically directed acceleration components in the problem's figure.
5. Solve θ̈ and check direction. A positive/negative sign must be interpreted using the given fork rotation convention.

2.9 Belt, wheel, and pulley points — the no-slip meaning

If a belt travels over a pulley without slipping, the **tangential speed of the rim equals the belt speed** at their contact. A point on the pulley experiences both inward normal acceleration and tangential acceleration; a particle on a straight section of belt has *no curvature acceleration*.

FIXED-RADIUS ROTATION

$$v = R\omega, \qquad a_n = R\omega^2 = \frac{v^2}{R}, \qquad a_t = R\alpha$$

R = pulley radius (m), ω = angular speed (rad/s), α = angular acceleration (rad/s²), a_n = inward acceleration (m/s²), a_t = along belt or rim tangent (m/s²).

How to solve Tutorial 2 Q5: recover ω from belt speed and radius at the instant. The total acceleration at a rim point is the perpendicular combination $\sqrt{(a_n^2+a_t^2)}$; use the stated magnitude to recover α. At another rim radius compute both components again; at a point on a straight belt only tangential acceleration remains. The tutorial gives the check answers **α = 300 rad/s²; a_B = 37.5 m/s²; a_C = 22.5 m/s²**. Radius values must be read from the tutorial diagram.

2.10 Relative motion — understand what the slash means

"A/B" always means "A as seen from B", not A divided by B. Imagine standing on B and looking at A: its observed velocity is the difference between their absolute velocities. The Week 2 lecture uses **translating, non-rotating reference frames** for these equations.

RELATIVE MOTION OF A AND B

$$\vec{r}_A = \vec{r}_B + \vec{r}_{A/B}$$

$$\vec{v}_{A/B} = \vec{v}_A - \vec{v}_B\,, \quad \vec{a}_{A/B} = \vec{a}_A - \vec{a}_B$$

Use x and y components separately. If the observer's frame rotates, these simple derivative relations no longer account for all rotating-frame terms.

Why this is useful: in a car overtaking another car, the closing speed is their velocity difference. With two perpendicular motions, subtract vector components first, then use Pythagoras. Do not subtract speed magnitudes unless they are already collinear.

Week 3 — Plane kinetics: connect forces to acceleration

Source focus: Week 3 pp. 3–25; Tutorial 3 questions 1–5.

3.1 Newton's second law and the free-body diagram

$\Sigma\vec{F}=m\,\vec{a}$; $\Sigma F_x=m\,a_x$; $\Sigma F_y=m\,a_y$.

Newton's second law is a **vector** equation. A free-body diagram (FBD) isolates one object and shows only the real forces applied by other bodies. In a kinetic diagram, draw the product ma separately to match the equations. The lecture also presents D'Alembert's dynamic equilibrium form $\Sigma\vec{F}-m\vec{a}=0$, with an auxiliary inertia force $-m\vec{a}$.

Method you should write in the exam

1. Isolate the particle or system. Draw weight mg vertically downward, surface normals, applied forces, spring force, cable tension and actual friction directions.
2. Choose x–y or n–t or r–θ axes. Indicate assumed positive acceleration direction.
3. Resolve every applied force into the chosen axes; write ΣF=m a component-wise.
4. Solve unknown contact reactions, tensions and acceleration. If speed/time are requested, follow with kinematics or energy.

Force	Magnitude/direction	Key warning
Weight	W=mg vertically downward	In lb-force problems mass=W/g in slugs.
Normal	N perpendicular to smooth surface	Not necessarily equal to mg.
Sliding friction	$F_k=\mu_k N$, opposing sliding	Use normal first, then friction.
Ideal spring	$F_s=k(L-L_0)$, restorative	Spring extension changes with geometry.
Tension	Along taut cord, pulling away	System pulley may impose velocity ratios.

3.2 Cartesian kinetics and angled applied forces

For a crate pulled by an inclined force, do **not** directly set N=mg. Upward components of pull reduce N. Calculate friction only after N is known.

Worked model 8 — Week 3 lecture crate: force then kinematics

50-kg crate, $\mu_k=0.3$, 400-N pull at 30° above the horizontal, initially at rest.

> $\Sigma F_y=0$: $N-50(9.81)+400\sin30°=0 \Rightarrow$ **N=290.5 N**.
> $\Sigma F_x=ma$: $400\cos30°-0.3(290.5)=50a \Rightarrow$ **a=5.185 m/s²**.
> $v=v_0+at=0+5.185(3)=$**15.6 m/s** after 3 s.

Answer pattern: FBD → y-equilibrium → friction → x-motion → kinematics. The lecture sometimes labels acceleration with m/s instead of m/s²; acceleration unit must be m/s².

Worked model 9 — Tutorial 3, Q1: spring force changes with position

A 2-kg smooth collar slides on a vertical guide, attached to a spring with k=3 N/m and natural length 0.75 m. Find instant acceleration and guide reaction 1 m below its initial position.

The spring initially spans 0.75 m horizontally. New length $L=\sqrt{(1^2+0.75^2)}=1.25$ m, extension s=0.50 m, $F_s=ks=1.50$ N. Its unit direction toward support has horizontal ratio 0.75/1.25 and vertical ratio 1/1.25.

> Horizontal: $N=1.5(0.75/1.25)=$**0.900 N**.
> Taking downward positive: $2a=19.62-1.5(1/1.25)=18.42 \Rightarrow$ **a=9.21 m/s² downward**.

Why not constant-acceleration formulas? As the collar moves, spring length and its angle change. The answer gives acceleration at *one instant*, not a constant value over the path.

3.3 Motion on a curved guide: normal and tangential force equations

> $\Sigma F_n=m(v^2/\rho)$; $\Sigma F_t=m(dv/dt)$.

Always draw the inward normal. For a roller over a hill, it often points downward toward the curve center; for a body at the bottom of a valley, it points upward. The normal reaction and gravity must

produce the required **net** inward force; no new “centripetal force” should be inserted on the diagram.

If the path has equation y(x), compute curvature via $\rho=[1+(y')^2]^{3/2}/|y''|$. For an instantaneous speed supplied in the question, $a_n=v^2/\rho$ immediately follows. If a_t is also given, use $\Sigma F_t=ma_t$ for tangential friction.

Extra tutorial walkthrough — Tutorial 3, Q2: normal reaction on a ski-jump curve

A 150-lb skier reaches the bottom of a parabolic ski-jump profile with speed 65 ft/s. The diagram gives the vertex profile $y=x^2/200-200$ (feet).

The vertex has $y'=0$ and $y''=1/100$ ft^{-1}, so local curvature radius $\rho=[1+(0)^2]^{3/2}/(1/100)=100$ ft. The normal points up toward the local center; tangential slope is zero here.

> $a_n=v^2/\rho=65^2/100=$**42.25 ft/s² upward**.
> ΣF_n: $N-150=(150/32.2)(42.25)\Rightarrow$ **N≈347 lb**.

Why this works: Find curvature first, then apply $\Sigma F_n=ma_n$. The support force is not equal to body weight while the path curves.

3.4 Kinetics in polar axes

> $\Sigma F_r=m(\ddot{r}-r\omega^2)$; $\Sigma F_\theta=m(r\alpha+2\dot{r}\omega)$.

The first step is **geometry**. In the Week 3 slotted-arm example the collar location is determined by $r=0.4\csc\theta$. Differentiate this, obtain radial and transverse accelerations, and then resolve real forces onto u_r and u_θ. A smooth slot acts **normal to the slot**, while a smooth curved guide acts normal to the curve — these are not necessarily radial.

Worked model 10 — Tutorial 3, Q3: smooth rotating arm with circular guide

0.5-kg collar; $r=0.8\cos\theta$ m, $\theta=45°$, $\dot{\theta}=3$ rad/s constant.

From the geometric constraint: $r=0.5657$ m, $\dot{r}=-0.8\sin45°(3)=-1.697$ m/s and $\ddot{r}=-0.8\cos45°(3^2)=-5.091$ m/s².

> $a_r=\ddot{r}-r\omega^2=-5.091-0.5657(9)=$**−10.18 m/s²**.
> $a_\theta=r\alpha+2\dot{r}\omega=0+2(-1.697)(3)=$**−10.18 m/s²**.
> ΣF_r: $-N\cos45°=0.5a_r\Rightarrow$ **N=7.2 N**.
> ΣF_θ: $F-N\sin45°=0.5a_\theta\Rightarrow$ **F=0 N**.

Why include the force directions? The circular guide normal and slot-arm contact enter different radial and transverse components. A “guess the force is radial” shortcut would fail.

Extra tutorial walkthrough — Tutorial 3, Q4: baggage truck and coupling

A 900-lb truck pulls 550-lb and 325-lb carts on level ground. Driving force F=40t lb; start from rest. Find speed at t=2 s and coupling force between truck and first cart.

Total weight is 1775 lb, so total inertial mass is 1775/32.2 slugs. For the whole system F=Ma, a(t)=(40t)(32.2)/1775=0.7256t ft/s².

> v(2)=$\int_0^2$0.7256t dt = **1.45 ft/s**.
> At t=2: a=1.451 ft/s². Isolate carts B+C (875 lb):
> T=(875/32.2)(1.451)=**39.4 lb**.

Strategy: the entire system eliminates internal couplings for acceleration. Isolate only the carts to obtain the force in the coupling.

Extra tutorial walkthrough — Tutorial 3, Q5: skateboarder on a circular track

A 60-kg skateboarder starts from rest on a circular track of radius 4 m, then reaches θ=60° (angle measured as shown in the tutorial diagram).

For motion along a circular arc, a_t=g cosθ and ds=R dθ, so v dv = Rg cosθ dθ. Integrate from θ=0, v=0.

> v²/2=Rg sin60° ⇒ v²=2(4)(9.81)sin60°=**67.97 m²/s²**.
> ΣF_n: N−mg sin60°=mv²/R.
> N=60(67.97/4)+60(9.81)sin60°=**1.53 kN**.

Key sequencing: tangential equation gives speed along the path; normal equation then gives reaction at the specified angle.

3.5 A system of particles

> $\Sigma \vec{F}_{external} = M\,\vec{a}_G$, where $M=\Sigma m_i$.

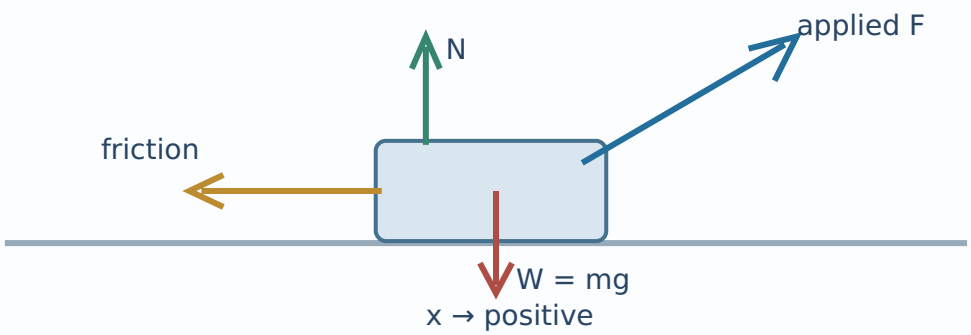
If several objects move together, treating them as one system cancels forces between them. Use a **separate subsystem FBD** when a coupling force is sought. This is the strategy used for the baggage truck and connected carts in Tutorial 3, Q4.

> **Exam remark.** Use force–acceleration for “find normal, tension, friction, acceleration now.” Use work–energy when the problem asks “speed after traveling from A to B.”

3.6 A genuine free-body diagram: what belongs and what does not

Isolate the object first. Draw a separate arrow for every **real external force acting on it**, with arrows originating at the particle model. The kinetic diagram is separate: it displays the direction and components of m a, which is a result of forces, not an additional physical force.

Object interaction	Correct force on the isolated object	What to check
Earth pulls particle	Weight W = mg, vertically downward	Not necessarily parallel to the incline.
Contact with smooth guide	Normal reaction N, perpendicular to local surface	No friction if “smooth” is specified.
Rough surface while sliding	Kinetic friction $F_k = \mu_k N$, opposite relative sliding	Direction is not always opposite the global x-axis.
Light taut cable	Tension T along the cable, pulling away	Same T through ideal frictionless massless pulleys; moving pulleys feel multiple segments.
Spring	Restoring force of magnitude k|x|	x measured from *unstretched* length; spring can pull or push depending on extension/compression.
Slot or rod constraint	Guide reaction normal to the contact surface	Can have a negative scalar result if assumed direction was reversed.



A typical horizontal rough-surface FBD. The actual normal reaction changes with the vertical component of the applied force; it is not automatically equal to mg.

3.7 Why normal reaction and acceleration are different questions

Normal reaction exists to keep the particle on a specified path; the **normal acceleration** describes how the velocity direction is changing. For a circular track, $a_n = v^2/R$ always points toward the centre. The reaction may point toward or away from the centre depending on which side of the surface supports the body.

NEWTON'S LAW IN MOVING TANGENT–NORMAL DIRECTIONS

$$\sum F_t = m\frac{dv}{dt}, \qquad \sum F_n = m\frac{v^2}{\rho}$$

ΣF_t = force sum along the tangent; ΣF_n = force sum toward the local centre of curvature; ρ = radius of curvature; v = speed. The sign of every actual force component must follow the selected axes.

RADIUS OF CURVATURE FROM A GRAPH Y = F(X)

$$\rho = \frac{\left(1 + \left(\frac{dy}{dx}\right)^2\right)^{3/2}}{\left|\frac{d^2y}{dx^2}\right|}$$

dy/dx = slope of path; d^2y/dx^2 = change of path slope. At the top of a hill, the n-direction points downward toward the curve's centre, even though the road's normal reaction points upward.

Method for parabolic hill (Week 3, pp. 25–28):

1. Use y(x) to calculate slope y′ and curvature radius ρ at A.
2. From slope, construct a local tangent and a normal pointing toward the concave side of the path.
3. Given speed v and tangential acceleration a_t, compute $a_n = v^2/\rho$.
4. Place weight mg, normal reaction N, and tangential friction F on the FBD. Convert 0.8 Mg to 800 kg.
5. Project $\Sigma F_t = ma_t$ and $\Sigma F_n = ma_n$. The forces the *wheels exert on the road* are opposite to the road's forces on the car (Newton's third law).

Additional lecture walkthrough — rotating-platform disk and breaking cord

Week 3 pp. 16–19: 3 kg disk on a rotating platform, radius 1 m, $\mu_k = 0.1$, initially at rest; cord breaks at tension 100 N. Find time.

Think: friction causes tangential acceleration; cord tension supplies inward centripetal force. These are perpendicular; do not add them as scalars.

NORMAL, TANGENT, AND VERTICAL EQUATIONS

$$\sum F_b = 0: \; N = mg = 29.43 \text{ N}$$

$$\sum F_t = ma_t: \; \mu_k N = ma_t \Rightarrow a_t = 0.981 \text{ m/s}^2$$

$$\sum F_n = m\frac{v^2}{R}: \; 100 = 3\frac{v^2}{1}$$

So v = √(100/3) = 5.77 m/s. With constant tangential acceleration, v = a_tt gives t = 5.77/0.981 = **5.89 s**. This is the lecture's solution structure.

Core lesson: the centripetal equation determines the *breaking speed*; the tangential equation determines *how long it takes to reach it*.

3.8 Recognise constraints before writing equations

With two bodies connected by an ideal cord, draw **separate FBDs**. Their acceleration magnitudes may differ when a moving pulley changes the cable length. Derive the cable constraint from the total length before claiming $v_A = v_B$. For a system of particles, internal cable tensions cancel only when you include both interaction partners inside the chosen system.

Common spring-collar trap: calculate the spring's *current length* using the given geometry, then subtract its unstretched length to find extension. Do not use the collar's vertical travel as the spring extension unless the spring lies along that same line.

Week 4 — Work, energy, power and efficiency

Source focus: Week 4 pp. 3–31 and Tutorial 4 questions 4–5 (as supplied).

4.1 Work is force acting through displacement

$dU=\vec{F}\cdot d\vec{r}=F\cos\varphi\, ds$; $U_{1\to2}=\int\vec{F}\cdot d\vec{r}$.

Work is a **scalar**, even when its force and displacement are vectors. Positive work adds kinetic energy; negative work removes it. For a constant force at constant angle, $U=F(s_2-s_1)\cos\varphi$. For a variable force integrate its component along the motion. The signed area under an F–s graph equals work.

Force type	Work from 1 to 2	Interpretation
Constant applied F	$F\Delta s\cos\varphi$	Angle to motion matters.
Weight	$mg(y_1-y_2)$	Downward displacement gives + work.
Ideal spring	$\frac{1}{2}k(s_1^2-s_2^2)$	Deforming a spring costs energy.
Sliding friction	$-\int\mu_k N\, ds$	Usually dissipative and negative.
Normal reaction (stationary smooth surface)	0	Perpendicular to motion.

4.2 Work–energy theorem

$T=\frac{1}{2}mv^2$; $T_1+\Sigma U_{1\to2}=T_2$.

The theorem comes from $\Sigma F=ma$ and $a=v\,dv/ds$. Force work accumulated over the path equals the change in kinetic energy. This bypasses time and is very effective for complicated variable forces.

Method you should write in the exam

1. State initial and final positions and identify which bodies are in the system.
2. Draw FBDs; list each force that does work, including sign and distance.
3. Write initial and final kinetic energy for each moving mass.
4. Apply $T_1+\Sigma U=T_2$ with all cable displacement/velocity constraints substituted.
5. If time is separately required, use $v=ds/dt$ to integrate after obtaining v(s).

Worked model 11 — Tutorial 4, Q4: pulley displacement ratio matters

Masses A=10 kg and B=100 kg connected with a cord through two movable pulleys on B; from rest, find B displacement when speed B reaches 2 m/s.

Cable constraint from geometry: $s_A+4s_B=$constant, so $\Delta s_A=-4\Delta s_B$ and $v_A=-4v_B=-8$ m/s (downward positive for both).

> $0+(98.1)(\Delta s_A)+(981)(\Delta s_B)=\frac{1}{2}(10)(8^2)+\frac{1}{2}(100)(2^2)$.
> Substitute $\Delta s_A=-4\Delta s_B$: $(-392.4+981)\Delta s_B=520$.
> **$\Delta s_B=0.883$ m downward.**

Why the minus sign? A goes upward while B descends. The pulley's four supporting strands create a 4:1 speed ratio. The support reactions do no work because their points of application are fixed.

Lecture expansion — Week 4, horizontal push with inclined spring

A 10-kg block moves 2 m up a smooth 30° incline under a horizontal force P=400 N. A spring (k=30 N/m) starts stretched 0.5 m and ends stretched 2.5 m. Find total work.

List the work from *each* force; take upward slope travel as positive. The incline reaction is perpendicular to displacement, so it does no work.

> $U_P=400(2\cos30°)=+692.8$ J.
> $U_{spring}=\frac{1}{2}(30)(0.5^2-2.5^2)=-90.0$ J.
> $U_{weight}=-(10)(9.81)(2\sin30°)=-98.1$ J; $U_N=0$.
> **$\Sigma U\approx504.7$ J (≈505 J)**.

Why individual terms? Work is scalar. Distinct forces add algebraically, and their signs communicate whether they help or resist the motion.

Lecture expansion — Week 4, braking distance on a downhill slope

A 3500-lb car travels downhill at 20 ft/s on a 10° road and skids with $\mu_k=0.5$. Find distance to rest.

Normal reaction $N=3500\cos10°=3446.8$ lb; friction magnitude is 1723.4 lb uphill. Weight performs positive work downhill; friction performs negative work.

> $\frac{1}{2}(3500/32.2)(20^2)+[3500\sin10°-1723.4]s=0$.
> $21739-1115.6s=0 \Rightarrow$ **$s\approx19.5$ ft downhill**.

Why work–energy? The question asks for a distance rather than time; start and end speeds are known.

4.3 Variable force and finding time afterwards

If a winch exerts F(s), the work is $\int F(s)ds$, giving a relation v(s). To obtain time, use $t=\int ds/v(s)$. This is why the Week 4 crane example solves for final speed *before* integrating for time.

Lecture expansion — Week 4, crane with force varying by height

A crane lifts a 2.50-Mg beam from rest with $F(s)=(28+3s^2)$ kN while raising it to s=3 m. Find final speed and elapsed time.

Work depends on position, so integrate before finding speed. The beam weighs $2500(9.81)=24.525$ kN downward.

> $\frac{1}{2}(2500)v^2=\int_0^s(28+3u^2)1000du-24525s = 3475s+1000s^3$.
> $v(s)=\sqrt{(2.78s+0.8s^3)}$. At s=3, **v=5.47 m/s**.
> $t=\int_0^3 ds/\sqrt{(2.78s+0.8s^3)}\approx$**1.79 s**.

Why two stages? Energy provides v(s); time is recovered from dt=ds/v(s), rather than assuming constant acceleration.

4.4 Power and efficiency

> $P=dU/dt=\vec{F}\cdot\vec{v}$; $\eta=P_{out}/P_{in}$; 1 hp=550 ft·lbf/s≈746 W.

Instantaneous power depends on the force **and the velocity of its application point**. For a known motor output P and speed v of the cable, the pulling force is F=P/v (assuming force parallel to motion). Mechanical efficiency is useful output divided by input; motor input is output divided by η.

Worked model 12 — Tutorial 4, Q5: winch, friction and changing power

800-lb log pulled at constant 4 ft/s up a 30° slope with 6 hp output. Find μ_k. Then output rises to 8 hp at that same instant; find acceleration.

At constant speed, a=0. Normal N=800 cos30°=692.82 lb. Winch pull is P/v.

> $F_6=(6\times550)/4=825$ lb.
> $0=825-800\sin30°-\mu_k(692.82)\Rightarrow$ **μ_k=0.613**.
> $F_8=(8\times550)/4=1100$ lb; net upslope force=1100−400−425=275 lb.
> Mass=800/32.2=24.84 slugs ⇒ **a=11.1 ft/s² upslope**.

Why is the speed still 4 ft/s? Acceleration changes instantly when power changes; velocity cannot jump instantaneously. Use the pre-change speed to compute the immediate force.

> **Exam remark.** The Week 4 supplied tutorial file contains specifically Q4 and Q5, not Q1–Q3; this guide uses the supplied solutions and the lecture's other examples for full chapter coverage.

4.5 A complete work-sign guide: which forces transfer energy?

Work is a dot product. The force direction relative to *actual displacement* determines its sign, not whether the force arrow looks upward or downward on the page. A normal contact force does no work in an ideal fixed smooth guide because displacement is tangent and reaction is perpendicular at every instant.

WORK OF A FORCE AND THE AREA IDEA

$$dU = \vec{F} \cdot d\vec{r} = F\cos\phi\, ds$$

$$U_{1\to 2} = \int_{s_1}^{s_2} F_t(s)\, ds$$

U = work (J = N·m); F_t = force component along displacement (N); ds = small path length (m); φ = angle between force and displacement. A force–displacement graph's signed area equals work.

Force	Work for motion 1→2	Reason for sign
Constant applied F at fixed φ	$U = F(s_2 - s_1)\cos\varphi$	Positive if along displacement, negative if opposing.
Gravity	$U_g = mg(y_1 - y_2)$	Positive for falling, negative for climbing, y positive upward.
Linear restoring spring	$U_s = \frac{1}{2}k(x_1^2 - x_2^2)$	Spring does negative work when deformation grows.
Sliding friction	$U_f = -\int \mu_k N\, ds$	Negative for sliding against friction; simplify to $-\mu N s$ if N constant.
Ideal fixed-guide normal force	$U_N = 0$	Force perpendicular to displacement.

When to choose energy instead of Newton's law: if the problem states an initial speed, a final height/position and asks for another speed, energy often bypasses the unknown time and acceleration history. If the problem asks for a contact reaction *at that final position*, use energy to find v there and then Newton's law to find the reaction.

4.6 Work–energy for a connected system: cable geometry first

In Tutorial 4 Q4, the four rope segments support the movable block, so a change in the blocks' positions obeys $\mathbf{\Delta s_A + 4\Delta s_B = 0}$, where both position coordinates were chosen positive downward. Differentiate to obtain $v_A = -4v_B$. Thus when B travels down by s, A moves up by 4s; the gravitational work of A is negative while the gravitational work of B is positive.

SOLVE CABLE LENGTHS BEFORE ENERGY

$$s_A + 4s_B = \text{constant} \quad \Longrightarrow \quad v_A + 4v_B = 0$$

$$\frac{1}{2}m_A v_{A1}^2 + \frac{1}{2}m_B v_{B1}^2 + U_g = \frac{1}{2}m_A v_{A2}^2 + \frac{1}{2}m_B v_{B2}^2$$

m_A, m_B = masses; v_A, v_B = signed velocities; U_g = total work of gravity. The tutorial obtains $s_B \approx 0.883$ m downward.

4.7 How variable-force problems can ask for BOTH speed and time

If force varies with displacement, integrating the force gives work and therefore speed as a function of position. **That does not directly give time.** Once v(s) is known, return to v = ds/dt and integrate dt = ds/v(s). Week 4's crane example deliberately tests this two-stage idea.

TWO-STAGE STRATEGY

$$\frac{1}{2}mv(s)^2 - \frac{1}{2}mv_0^2 = \int_{s_0}^{s} F_{\text{net}}(u)\,du$$

$$t - t_0 = \int_{s_0}^{s} \frac{du}{v(u)}$$

Use a dummy variable u inside the integral. First find speed at the requested distance; then integrate reciprocal speed to obtain elapsed time. Variable force means variable acceleration, so constant-a shortcuts are generally invalid.

4.8 Power and efficiency: a source of easy lost marks

AT AN INSTANT, FORCE AND VELOCITY MUST BE PAIRED

$$P = \vec{F}\cdot\vec{v} = Fv\cos\phi\,, \qquad \eta = \frac{P_{\text{useful out}}}{P_{\text{in}}}$$

P = mechanical power (W); η = efficiency (dimensionless); φ = angle between force and velocity. Motor input power = useful output power divided by η, not multiplied by η.

Additional lecture walkthrough — power supplied by a pushing force

Week 4 pp. 26–28: 50 kg crate, force 150 N at a 3–4–5 downward angle, μ_k = 0.2, starts at rest. Find supplied power at t = 4 s.

Why this order? Power needs instantaneous velocity. Velocity needs acceleration. Acceleration needs the FBD and the normal reaction.

Step 1. Resolve F_x = (4/5)(150) = 120 N, downward F_y = (3/5)(150) = 90 N.

Step 2. Vertical equilibrium gives N = mg+90 = 580.5 N. Friction = 0.2N = 116.1 N.

Step 3. Horizontal ΣF = ma: a = (120−116.1)/50 = 0.078 m/s². At t = 4 s, v = at = 0.312 m/s.

Step 4. Only horizontal force component does work: P = F_xv = 120(0.312) = **37.4 W** (rounded).

Check: P ≠ 150v, because the applied force is angled while motion is horizontal.

Week 5 — Potential energy and conservation of mechanical energy

Source focus: Week 5 pp. 3–15; Tutorial 5 question 1 and the momentum problems in questions 2–3.

5.1 Conservative versus nonconservative forces

Conservative work depends only on end positions; the work equals a drop in potential energy. Ideal weight and spring forces are conservative. Kinetic friction is nonconservative and converts mechanical energy into heat. A chosen datum determines the numerical gravitational potential, but not the energy difference.

> $V_g = mgy$; $V_s = \frac{1}{2}ks^2$; $V = V_g + V_s$; $U_{conservative,1\to2} = V_1 - V_2$.

The spring deformation s can be positive (stretching) or negative (compression); s^2 makes elastic potential energy nonnegative. For an angled spring, first calculate actual length L from geometry, then $s = L - L_0$. For multiple springs, use a separate $\frac{1}{2}k_i s_i^2$ for each spring in contact.

5.2 The conservation equation and how to avoid double counting

> General: $T_1 + V_1 + U_{nonconservative} = T_2 + V_2$. Without nonconservative work: $T_1 + V_1 = T_2 + V_2$.

This is the Week 4 work–energy theorem reorganized by replacing gravitational and spring work with changes of potential energy. Either use **work by weight/spring** or **their potential energy**; never both in the same balance.

Method you should write in the exam

1. Choose two positions, a vertical datum, and a positive height direction.
2. Write the kinetic and potential energies at both states (including every spring).
3. Calculate any nonconservative work, such as $-\mu Ns$.
4. Relate motions of multiple bodies using cable-length constraints if present.
5. Solve the energy equation; reject geometrically impossible roots. If forces at final point are asked, use Newton separately.

Worked model 13 — Tutorial 5, Q1: vertical collar with horizontal spring

2-kg collar initially has an unstretched horizontal spring of natural length 0.75 m, k=3 N/m. Collar descends 1 m. Find speed if it begins (a) at rest; (b) with upward speed 2 m/s.

The final spring length is 1.25 m, hence stretch s=0.50 m. Final stored energy is $\frac{1}{2}(3)(0.5^2) = 0.375$ J. Vertical drop releases $mg(1) = 19.62$ J.

> (a) $0 + 19.62 = \frac{1}{2}(2)v^2 + 0.375 \Rightarrow$ **$v = 4.39$ m/s**.
> (b) $\frac{1}{2}(2)(2^2) + 19.62 = \frac{1}{2}(2)v^2 + 0.375 \Rightarrow$ **$v = 4.82$ m/s**.

Why is upward starting velocity still positive energy? Kinetic energy depends on speed squared. The direction at the initial instant does not make kinetic energy negative.

5.3 Combining energy with instantaneous contact force

In the Week 5 gantry-airplane lecture example, gravitational potential loss yields speed at the second angle. Then $\Sigma F_n = mv^2/\rho$ gives cable tension. Energy alone cannot directly determine that reaction. Similarly, a skateboarder on a circular track may require speed from energy and normal force from Newton's second law.

Lecture expansion — Week 5, airplane gantry: energy then tension

An 8-Mg airplane suspended on a 20-m cable is released from rest at θ=60° and moves to θ=15°. Find speed and supporting cable tension at 15°.

Take the height reference at the support. Energy loss is mgL(cos15°−cos60°); a tangential speed can be obtained without evaluating time.

> ½(8000)v²=(8000)(9.81)(20)(cos15°−cos60°) ⇒ **v=13.52 m/s**.
> T−mg cos15°=mv²/20 ⇒ **T≈149 kN**.

Two laws for two outputs: gravitational energy supplies speed; the radial Newton equation supplies the force.

5.4 Maximum spring compression: solve for geometry, not time

At a maximum compression, the moving ram has v=0. Calculate gravity's drop from start to that compressed position and energy in **each active spring**. If nested springs start touching at different positions, the second spring deformation is $s_B=\max(0,s_A-\text{gap})$. The lecture's Week 5 example uses a positive quadratic root to find compression, then confirms both springs are active.

> **Exam remark.** In Week 5 lecture pp. 10 and 13, a few rendered equations have inconsistent typography; use the underlying relations T=½mv² and $a_n=v^2/\rho$. In Week 5 p. 15 the stiffness term should follow the stated spring stiffness $k_A=12$ kN/m.

Lecture expansion — Week 5, ram compressing two nested springs

A 100-kg ram drops from rest 0.75 m above spring A (k_A=12 kN/m). Spring B has k_B=15 kN/m and engages after 0.10 m compression of A. Find the maximum A compression s.

At maximum compression speed is zero. The ram descends 0.75+s. For a physically valid two-spring state, s≥0.10 m and spring B compression is s−0.10.

> ½(12000)s²+½(15000)(s−0.1)²=981(0.75+s).
> 13500s²−2481s−660.75=0.
> Positive physical root: **s_A=0.331 m**, **s_B=0.231 m**.

Why check the root? An algebraic root is only valid if the geometry and spring-engagement assumption are satisfied.

5.5 How potential energy simplifies everything

Conservative force work depends only on initial and final positions. You can record gravity and spring effects as **changes in stored energy**, instead of separately adding their force work. This is the main benefit of using a total potential $V = V_g + V_e$.

MECHANICAL ENERGY WITH OPTIONAL NONCONSERVATIVE WORK

$$T_1 + V_1 + U_{1\to 2}^{\mathrm{NC}} = T_2 + V_2$$

$$T = \frac{1}{2}mv^2, \quad V_g = mgy, \quad V_e = \frac{1}{2}kx^2$$

T = kinetic energy (J), V = potential energy (J), U^{Nc} = work from nonconservative forces (J), y = height above chosen datum (m), x = compression/extension relative to unstretched spring (m). When $U^{Nc} = 0$, total mechanical energy is constant.

Situation	What changes	How you record it
Object descends	Height y decreases	V_g decreases → T may increase.
Spring compresses	|x| increases	V_e increases → T may decrease.
Object crosses datum	y changes sign	V_g may be negative: no issue.
Sliding friction acts	Mechanical energy dissipated	$U^{Nc} = -\mu Ns$ for constant N.
Maximum compression	Momentary speed is zero	Set $T_2 = 0$, but a_2 need not be zero.

Two ways to count gravity or spring, never both: (A) $T_1 + U_{gravity} + U_{spring} + U_{other} = T_2$, or (B) $T_1 + V_1 + U_{other} = T_2 + V_2$. Double-counting is a very common error.

5.6 Nested springs: understand when the second spring engages

In the Week 5 ram example, the ram falls 0.75 m before contacting spring A. Spring B is 0.10 m shorter, so it does **not** begin compressing until spring A has already compressed 0.10 m. At maximum downward motion, $x_A = s$ and $x_B = s - 0.10$ (only if $s \geq 0.10$). Write potential energy using the same final position of the ram in every term.

ENERGY BALANCE FOR MAXIMUM RAM COMPRESSION

$$mg(0.75 + s) = \frac{1}{2}k_A s^2 + \frac{1}{2}k_B(s - 0.10)^2$$

m = 100 kg; g = 9.81 m/s²; k_A = 12,000 N/m; k_B = 15,000 N/m; s = compression of spring A measured downward (m), so spring B compresses s − 0.10 m. This only applies when $s \geq 0.10$ m.

Step-by-step solving:

1. Choose datum at ram's starting height. It starts from rest, $T_1 = 0$.
2. At maximum compression, $v_2 = 0$. Gravitational energy lost over 0.75+s becomes stored spring energy.
3. Substitute k in N/m, not kN/m. Expand and solve the quadratic.
4. Keep the physically allowed positive root $s \geq 0.10$ m; reject roots implying penetration or incorrect spring engagement.
5. The lecture obtains $s_A = 0.331$ m and $s_B = 0.231$ m.

Slide-reading caution: the extracted formula text on Week 5 p. 15 can display a stiffness coefficient as “1200” due to a formatting loss. Use the stated k_A = 12 kN/m = 12,000 N/m. The displayed quadratic and final result are consistent with 12,000 N/m.

5.7 Energy gives speed; a force equation gives tension

A classic two-stage problem is a particle moving on a pendulum or curved support. Energy uses height change to find speed at a new position; the instantaneous n-direction law determines tension. Neither can replace the other if both speed and tension are asked.

CURVED SUPPORT AT ANGULAR POSITION B

$$v_2^2 = v_1^2 + 2g(y_1 - y_2)$$

$$\sum F_n = \frac{mv_2^2}{R}$$

At the chosen location resolve weight and tension along the *actual* inward normal. R = radius of path. For the Week 5 airplane gantry (8 Mg on a 20 m cable), the source gives v ≈ 13.52 m/s and maximum tension ≈ 149 kN at the stated lower position.

Week 6 — Impulse, momentum, collisions and systems

Source focus: Week 6 pp. 3–15; Tutorial 5 pp. 1 and 3–6; Tutorial 6 pp. 1–5.

6.1 Linear momentum and linear impulse

$\vec{p}=m\vec{v}$; $\vec{I}=\int_{t_1}^{t_2}\vec{F}\,dt$; $m\vec{v}_1+\Sigma\vec{I}=m\vec{v}_2$.

Momentum is a vector measured in kg·m/s. Impulse is the area under the force–time graph, in N·s. The impulse–momentum law is Newton’s second law integrated over time; it works for varying force, not only constant force. If force is constant, I=FΔt.

When time is specified but displacement is not, impulse–momentum is often shorter than work–energy. You must still draw the FBD and include all relevant force impulses, including gravitational and support impulses where they are significant.

Method you should write in the exam

1. Choose a time interval and sign convention. Draw the initial and final momentum vectors.
2. Isolate the particle and draw all forces that act through the interval.
3. Compute their impulses ∫Fdt, taking signs and components.
4. Write $mv_1+\Sigma I=mv_2$ in x and y separately; solve for final v or reaction.

Worked model 14 — Week 6 lecture: constant pulling impulse

100-kg stone, initially at rest on a smooth horizontal floor, pulled by 200 N at 45° above horizontal for 10 s.

> Horizontal: $0+(200\cos 45°)(10)=100v_{x2} \Rightarrow \mathbf{v_{x2}=14.1\ m/s}$.
> Vertical (no vertical motion): $0+[N-981+200\sin 45°](10)=0 \Rightarrow \mathbf{N\approx 840\ N}$.

Why use two impulse equations? The horizontal impulse causes a momentum increase; vertically, impulses balance and the stone remains supported.

Lecture expansion — Week 6, trampoline safety: gravitational speed then impulse

Ten people, each 50 kg, fall from 2 m above a trampoline and come to rest over 0.5 s with constant deceleration. The trampoline fails above 5000 N total force.

Before landing, speed from free fall is $v=\sqrt{2gh}$. During the 0.5-s deceleration, the trampoline pushes up while gravity acts down.

> $M=10(50)=500$ kg; impact speed $v=\sqrt{[2(9.81)(2)]}=6.26$ m/s downward.
> Taking upward positive, $a=[0-(-6.26)]/0.5=12.53$ m/s².
> $N-Mg=Ma \Rightarrow N=500(12.53+9.81)\approx \mathbf{11.2\ kN} > 5.0$ kN.

Conclusion: under the lecture assumptions the threshold is exceeded. The crucial distinction is that the surface force must both cancel weight and decelerate the moving group.

6.2 Conservation of total linear momentum

> $\Sigma m_i\vec{v}_{i,1}+\Sigma\vec{I}_{ext}=\Sigma m_i\vec{v}_{i,2}$. If $\Sigma\vec{I}_{ext}=0$, then $\Sigma m_i\vec{v}_{i,1}=\Sigma m_i\vec{v}_{i,2}$.

Internal impulses between bodies cancel when the whole system is chosen. During a short collision or explosion, external impulses may be negligible relative to the impact or explosion impulse. However, momentum is conserved **component by component** only if the net external impulse in that direction can be neglected.

Kinetic energy generally is **not** conserved in an explosion or inelastic impact. A mass center with no net external impulse continues with constant velocity: $M\vec{v}_G=\Sigma m_i\vec{v}_i$.

Worked model 15 — Tutorial 5, Q2: an exploding projectile in 3D

A 20-kg projectile explodes into fragments of masses 5, 9, 6 kg at the top of its trajectory. Known velocities/trajectories of two fragments allow the third velocity to be found.

Structure used in the supplied solution: first compute shell velocity at apex (vertical component zero), then use projectile motion to reconstruct fragment A and B velocities, then conserve vector momentum *during the short explosion*.

> $m\,\vec{v}_{before} = m_A\vec{v}_A + m_B\vec{v}_B + m_C\vec{v}_C$.
> Solving the component equations in the supplied tutorial gives:
> $\vec{v}_C=\mathbf{427i-173.4j-82.5k\ m/s}$, $|\vec{v}_C|\approx\mathbf{468\ m/s}$.

Why separate the before/after instant? Momentum conservation applies during the explosion, while gravity-driven kinematics is used before/after the event to find the needed velocities.

6.3 Forces that are impulsive versus non-impulsive

The tutorial distinguishes a high-magnitude force over a short period, such as impact force, from ordinary forces whose impulse in that brief interval is negligible. This is not an absolute classification: **a weight force has a measurable impulse over a long enough time**. Always state which impulses you neglect and why.

Extra tutorial walkthrough — Tutorial 5, Q3: two connected blocks by impulse

A=3 kg is suspended from a moving pulley; B=5 kg hangs from the free end. Released from rest, find B speed in 6 s. Cord and pulley are ideal.

Use downward as positive for both bodies. Cord length gives $2s_A+s_B=\text{constant}$ and $2a_A=-a_B$. The moving pulley gives upward force 2T on A.

> A: $3g-2T=3a_A$; B: $5g-T=5a_B$.
> Use $a_A=-a_B/2 \Rightarrow$ **a_B=5.97 m/s² down**, T≈19.2 N.
> $v_B=a_B(6)$=**35.8 m/s down**.

What the supplied tutorial emphasizes: treat the cable constraint, each block's impulse, and the pulley's force multiplier as one simultaneous system.

6.4 A force–time graph tells you a velocity change

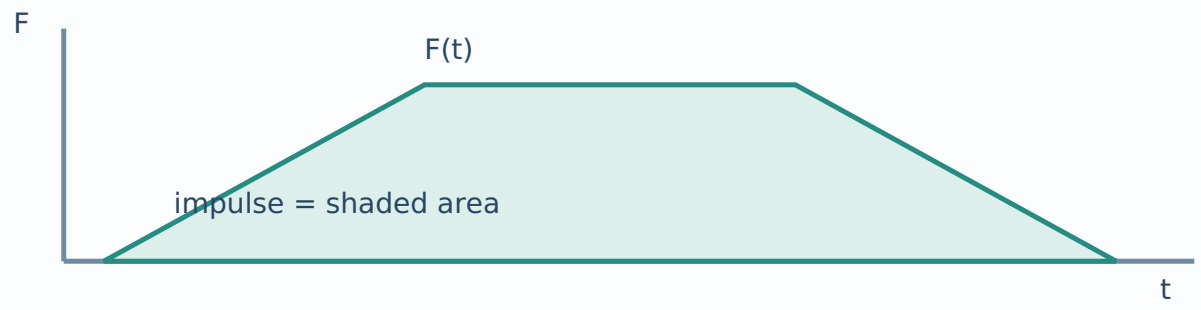
Work integrates force over **displacement**. Impulse integrates force over **time**. They have different units and answer different questions, so identify the horizontal axis of any graph before deciding which physical law it encodes.

FROM A FORCE HISTORY TO FINAL VELOCITY

$$\vec{I}=\int_{t_1}^{t_2}\vec{F}(t)\,dt=m(\vec{v}_2-\vec{v}_1)$$

$$\vec{F}_{\text{avg}}=\frac{\vec{I}}{t_2-t_1}$$

I = linear impulse (N·s), F_avg = average force (N), m = mass (kg), v = velocity vector (m/s). Signed area under an F(t) graph changes corresponding signed momentum.



Here the trapezoid's area is impulse, not work. If force reverses direction, its force–time area is negative in that direction.

6.5 How to inspect a system before saying momentum is conserved

Draw one boundary around all bodies that interact during the event. Forces between members are **internal** and cancel in the system total. The outside world can still apply external impulse. Conservation can hold in one coordinate direction but not another.

SYSTEM MOMENTUM EQUATION (COMPONENT-WISE)

$$\sum_i m_i \vec{v}_{i1} + \int_{t_1}^{t_2} \sum \vec{F}_{\text{ext}}\, dt = \sum_i m_i \vec{v}_{i2}$$

i labels each particle; m_i and v_i are its mass and velocity; F_ext includes only forces applied from outside the chosen system. If its impulse is negligible in x, conserve x-momentum even if y has significant external impulse.

Explosion method, exactly as Tutorial 5 Q2 requires:

1. Use projectile kinematics *before the explosion* to find velocity of the intact shell at point P.
2. Calculate the velocities of any fragments using their later trajectories. These are velocities *immediately after* the explosion, not their later impact speeds.
3. For the very short explosion, neglect non-impulsive external gravity relative to the explosion impulse and conserve momentum in x, y and z.
4. Write one vector equation $m\, v_{\text{before}} = m_A v_A + m_B v_B + m_C v_C$.
5. Solve the unknown fragment components first, then calculate its speed from the magnitude.

Diagnostic: kinetic energy is not in general conserved in an explosion; chemical energy may become kinetic energy. Linear momentum is conserved when net external impulse is negligible. This is why the tutorial uses momentum rather than setting initial and final kinetic energies equal.

6.6 Time-dependent applied force with friction

Week 6 lecture pp. 11–13 asks about a crate on an incline with P(t) = 20t lb. The applied force's impulse from 0 to 2 s is an integral, **not P(2) times 2**. First use the geometry and normal-force equation to obtain friction; then write the impulse equation along the incline.

ONE-DIMENSIONAL INCLINE TEMPLATE

$$m(v_2 - v_1) = \int_0^2 (P_{\parallel}(t) + W_{\parallel} - F_k)\, dt$$

$$\int_0^2 (20t)\, dt = 40 \text{ lb} \cdot \text{s}$$

$P_\parallel$ = applied-force component along the chosen incline direction; $W_\parallel$ = signed weight component; F_k = signed kinetic friction. The applied P may be angled in the figure, so check its direction before taking its full 40 lb·s as the along-slope impulse.

Watch the units: in FPS, a weight specified in lb is a force (lbf), so use m = W/g in slugs, with g ≈ 32.2 ft/s². In SI, kilograms already measure mass.

Tutorial 6 extension — Angular momentum and angular impulse

Source focus: Tutorial 6 pp. 1–5. This is tutorial content beyond the explicitly named Week 6 linear-momentum slide headings.

7.1 Angular momentum of a particle about a fixed point

$\vec{H}_O = \vec{r}_{O\to P} \times (m\vec{v})$; $\vec{M}_O = \vec{r}_{O\to P} \times \vec{F}$; $d\vec{H}_O/dt = \Sigma \vec{M}_O$.

The cross product measures the moment of linear momentum about reference point O. Its direction comes from the right-hand rule. The position vector must run **from O to the particle**. For motion in a plane about the perpendicular z-axis, $H_z = mr\, v_\theta = mr^2\omega$; the radial velocity component contributes zero.

If the resultant external angular impulse about O or the selected axis is zero: $H_1 = H_2$.

The important reasoning is to select the axis where external forces have zero moment. A cord pulling toward a fixed center produces **zero moment about that center** even if its tension changes significantly.

Worked model 16 — Tutorial 6, Q1: 3D cross products

$\vec{r} = (3i+6j+4k)$ m, $m = 2$ kg, $\vec{v} = 5j$ m/s and force $\vec{F} = 10k$ N.

$\vec{H}_O = \vec{r} \times (m\vec{v}) = (3i+6j+4k) \times (10j)$
$= \mathbf{-40i + 30k\ kg \cdot m^2/s}$.
$d\vec{H}_O/dt = \vec{M}_O = \vec{r} \times \vec{F} = (3i+6j+4k) \times (10k)$
$= \mathbf{60i - 30j\ N \cdot m}$.

Why cross products? The result cannot be found by simply multiplying position and speed as scalars in a 3D problem; the orientation matters.

Worked model 17 — Tutorial 6, Q2: angular momentum PLUS work-energy

0.8-lb ball on a smooth horizontal table, r_1=1.75 ft, initial transverse speed 4 ft/s; pulled toward the hole at 6 ft/s until r_2=0.6 ft.

The cord tension points through the reference axis, so it has zero moment there. Conserve z angular momentum to find the **transverse** final velocity; then add radial velocity to obtain total speed.

> $mr_1v_{\theta 1}=mr_2v_{\theta 2} \Rightarrow v_{\theta 2}=1.75(4)/0.6=$**11.67 ft/s**.
> $|v_2|=\sqrt{(11.67^2+6^2)}=$**13.1 ft/s**.
> Work by cord: $U=\frac{1}{2}(0.8/32.2)(13.1^2-4^2)\approx$**1.94 ft·lbf**.

Two different methods, two different questions: angular momentum finds the transverse component; energy finds work done by tension. The radial 6 ft/s makes no contribution to H_z.

7.2 Angular momentum with springs and bowl geometry

Tutorial 6 Q3 uses a disk attached to an elastic cord: because the central spring force has no moment about its anchor, conserve angular momentum for the transverse speed; use energy with $\frac{1}{2}ks^2$ to find total speed; then recover the radial speed from $v^2=v_r^2+v_\theta^2$. Tutorial 6 Q4 uses conservation of angular momentum about the vertical axis combined with gravitational energy loss: $v_\theta=r_0v_0/r$, $v=\sqrt{(v_0^2+2gh)}$, and the angle of velocity to the transverse horizontal tangent follows $\cos\theta=v_\theta/v$.

Extra tutorial walkthrough — Tutorial 6, Q3: disk with elastic central cord

A 2-kg disk on a smooth table starts with v_1=1.5 m/s tangent to an unstretched elastic cord of length r_1=0.5 m. k=20 N/m. Find its speed and rate of cord stretch at stretch 0.2 m.

At the second position r_2=0.7 m. Central tension has no moment about the anchor: $mr_1v_{\theta 1}=mr_2v_{\theta 2}$. Mechanical energy is also conserved.

> $v_{\theta 2}=(0.5)(1.5)/0.7=$**1.071 m/s**.
> $\frac{1}{2}(2)(1.5^2)=\frac{1}{2}(2)v_2^2+\frac{1}{2}(20)(0.2^2) \Rightarrow$ **v_2=1.360 m/s**.
> $\dot{r}_2=\sqrt{(v_2^2-v_{\theta 2}^2)}=$**0.838 m/s outward**.

Why both laws? Angular momentum determines the tangential velocity; energy determines the full speed; their difference in quadrature gives the radial stretching rate.

Extra tutorial walkthrough — Tutorial 6, Q4: moving inside a smooth hemispherical bowl

A particle enters a smooth hemispherical bowl tangent to its horizontal rim at radius r_0 with speed v_0. It later descends by h to radius r, with velocity angled θ from the horizontal tangent.

Weight and the bowl reaction have zero moment about the vertical axis, so that component of angular momentum is conserved. Gravity also gives the kinetic-energy increase.

> $mr_0v_0 = mr\ v\cos\theta$; $\frac{1}{2}mv^2 = \frac{1}{2}mv_0^2 + mgh$.
> By bowl geometry: $r^2 = r_0^2 - h^2$; $v = \sqrt{(v_0^2 + 2gh)}$.
> **$\theta = \cos^{-1}[r_0v_0/(\sqrt{(r_0^2 - h^2)}\sqrt{(v_0^2 + 2gh)})]$**.

Why this construction? The vertical axis is the special moment-free axis, while energy gives the new magnitude of velocity.

7.3 Cross products without guessing the sign

The order matters: r × F and F × r are opposites. Always draw r from the reference point to the particle first; then use the right-hand rule or the determinant formula. This avoids the most common sign mistakes in Tutorial 6 Q1.

CARTESIAN CROSS-PRODUCT EXPANSION

$$\vec{A} \times \vec{B} = (A_yB_z - A_zB_y)\vec{i} + (A_zB_x - A_xB_z)\vec{j} + (A_xB_y - A_yB_x)\vec{k}$$

A and B are any vectors with Cartesian components. For angular momentum, A = r and B = mv; for moment, A = r and B = F.

> **Quick memory check:** i × j = k; j × k = i; k × i = j. Reversing the order changes the sign. A radial velocity $v_r\ e_r$ gives zero moment about O because r and v_r are parallel.

7.4 Conservation is about the chosen axis, not automatically every component

ANGULAR IMPULSE–MOMENTUM ABOUT A FIXED POINT/AXIS

$$\vec{H}_{O2} - \vec{H}_{O1} = \int_{t_1}^{t_2} \sum \vec{M}_O\, dt$$

$$H_z = mr^2\dot{\theta} = mrv_\theta \quad \left(\sum M_z = 0 \Rightarrow H_{z1} = H_{z2}\right)$$

H = angular momentum (kg·m²/s); M = moment of external forces (N·m); r = distance from axis; vθ = perpendicular tangential component. The angular-momentum vector need not be entirely conserved merely because one axis has zero external moment.

Use angular momentum to obtain the new **tangential** speed when radial distance changes, and energy to obtain the **total** speed. In Tutorial 6 Q2, pulling the ball inward changes its speed because the cable does positive work even while having zero moment about the central axis. Zero torque does **not** mean zero work.

7.5 A two-law checklist for elastic cord or hemispherical bowl

1. Define the chosen central point or axis. Identify why the moment about it is zero: force points through the axis, or moment about vertical z is zero.

2. Conserve the **correct component** of angular momentum to find v_θ or the horizontal-tangent speed.
3. Use total energy with gravitational elevation and any elastic deformation to calculate the new $|v|$.
4. Recover the radial component from $v^2 = v_r^2 + v_\theta^2$, or an angle from the component ratio.
5. Check that the radical is non-negative. If geometry implies an impossible component larger than total speed, recheck the assumed reachable position.

How to construct full-mark exam answers

The tutorials repeatedly follow the same logical format. The marker should see not only your calculator result, but **why you selected that law**. Use the following answer skeleton.

Universal answer skeleton

1. **Given and required:** list quantities with SI/FPS units and mark the exact instant or interval.
2. **Diagram:** draw axes, position geometry, FBD, and/or velocity–momentum arrows. Label constraints.
3. **Governing principle:** write the main vector or scalar equation *before* inserting numbers.
4. **Special geometry:** spring length by Pythagoras, curvature ρ, polar $r(\theta)$, or constant cable length.
5. **Component equations:** show at least the x/y, n/t, r/θ or each momentum component required.
6. **Substitution:** use consistent positive directions and perform algebra with units.
7. **Interpretation:** present magnitude, direction and feasibility; check signs and limiting cases.

Common mistakes that cost marks

Mistake	Correction
$v=0$ at a turning point $\Rightarrow a=0$	Only v is zero; acceleration is generally not.
Using $s=v_0t+\frac{1}{2}at^2$ when a is changing	Integrate actual a(t), or use work–energy as appropriate.
Assuming normal force equals weight	Resolve all perpendicular force components first.
Assuming all acceleration points along velocity	Curved motion also has inward $a_n=v^2/\rho$.
Forgetting $2\dot{r}\omega$ in polar acceleration	Write both complete polar components.
Putting the spring/gravity into both U and V	Use either force work or conservative potential, not both.
Assuming equal speeds of pulley-connected blocks	Derive the constant-length relation first.
Using momentum conservation without checking outside forces	Check the external impulse direction by direction.
Confusing lb with slug or mm with m	Convert units before substituting.
Applying conservation of kinetic energy to collisions/ explosions	Momentum may be conserved but kinetic energy need not be.

Focused study order for a short preparation window

First: identify the axes and which principle is appropriate. **Second:** master the five tutorial patterns most likely to recur: 1D time functions, Cartesian/polar derivatives, FBD+$\Sigma F=ma$, spring/ pulley work–energy, and impulse/angular momentum. **Third:** reproduce complete tutorial solutions without looking at the answer and compare both the method and the final result.

Exam execution: a practical question-to-solution playbook

In the tutorial solutions the structure is remarkably repeatable. Do not jump from the diagram straight to numbers. First demonstrate that you know **which principle applies**, then make geometry and kinematics agree with that principle.

Playbook A — A force, tension, normal reaction or friction is requested

1. **Isolate the body** and label forces on a clean FBD; no m a arrow among the real forces.
2. **Pick coordinates** based on the geometry: Cartesian for straight rails; tangent–normal for a curve; polar for rotating/sliding mechanisms.
3. **Calculate acceleration** from the given motion or path first, including curvature where necessary.
4. **Write component equations** $\Sigma F_x=ma_x$, $\Sigma F_y=ma_y$ (or other axes).
5. **Solve forces** with the correct constraint equations, and interpret any negative sign.

Playbook B — A final speed after distance, spring compression or change in height

1. Choose the two positions (1 and 2). Write down v, y, spring deformation and any cable-related displacement in both states.
2. Decide whether to count gravity/spring as work or potential. Use only one representation.
3. Identify nonconservative work (especially friction or a motor).
4. Write $T_1 + V_1 + U^{Nc} = T_2 + V_2$, then substitute values.
5. If the problem subsequently asks force or time, use $\Sigma F=ma$ or $dt=ds/v(s)$ as the next stage.

Playbook C — A fast collision, explosion, or variable force lasting Δt

1. Choose the particle or whole interacting system and list external forces.
2. Write $m\,v_1 + \int F dt = m\,v_2$ or the corresponding system version *as a vector equation*.
3. Split x, y, and z, keeping velocity component signs.
4. Neglect an external force's impulse only when justified by the short event and force scale; do not assume “external” means negligible.
5. State the final velocity vector before taking the magnitude and direction.

Playbook D — An arm rotates or a cord is pulled toward a point

1. Check if r is constant. If r changes, don't delete $\dot{r}$, $\ddot{r}$ or the $2\dot{r}\dot{\theta}$ term.
2. When $r(\theta)$ is specified, use chain-rule derivatives before substituting numbers.
3. For forces on rotating slots, use polar $\Sigma F_r = ma_r$ and $\Sigma F\theta = ma\theta$.
4. For a central-force motion with moment zero about an axis, conserve angular momentum about *that axis*.
5. Combine with work–energy only when it supplies a different missing speed component or work quantity.

Master glossary of symbols, units and physical meanings

Symbol	Meaning	SI units	One reminder
s, x, y	Position coordinate, signed	m	Displacement is final minus initial.
r, $\vec{r}$	Radial distance / position vector	m	$\vec{r}$ points from reference origin to particle.
t, Δt	Time / elapsed time	s	Duration is positive.
v, $\vec{v}$	Signed component / vector velocity or speed by context	m/s	Speed is magnitude, never negative.
a, $\vec{a}$	Acceleration	m/s²	Can change speed or direction.
m, W	Mass / weight	kg / N	W = mg; 1 Mg = 1000 kg.
F, N, T	Applied force / normal / tension	N	Text uses T for kinetic energy in energy equations; context distinguishes them.
μ_k, k	Kinetic friction coefficient / spring stiffness	– / N/m	Spring force based on deformation.
θ, $\dot{\theta}$, $\ddot{\theta}$	Angle, angular velocity, angular acceleration	rad, rad/s, rad/s²	Use radians for differentiation.
$\dot{r}$, $\ddot{r}$	Rate of radius change / radial second derivative	m/s, m/s²	Not the full polar radial acceleration.
ρ	Radius of curvature	m	$a_n = v^2/\rho$ toward centre.
U, V, T_{kin}	Work, potential energy, kinetic energy	J	Choose a consistent bookkeeping convention.
P, η	Power / efficiency	W / –	Efficiency = useful out / supplied in.
I, $\vec{p}$, $\vec{H}$	Impulse, linear momentum, angular momentum	N·s, kg·m/s, kg·m²/s	Impulses integrate over time, moments use cross products.
i, j, k	Fixed Cartesian unit vectors	–	All mutually perpendicular.
e_r, eθ, e_t, e_n	Unit basis vectors for polar and path coordinates	–	They rotate as particle/axes turn.

> **FPS quick reference (when tutorials use lb and ft):** g ≈ 32.2 ft/s²; weight W in pounds-force corresponds to mass m = W/g in slugs; 1 hp = 550 ft·lbf/s. Always carry the unit through equations to catch confusion between mass and force.

Self-check before you call a topic understood

These are **retrieval prompts**, not worked examples; try them without opening the answers in the earlier chapters. A topic is understood when you can explain your method before looking up its equation.

Week 1: Given v(t) that crosses zero, explain how to find distance and displacement separately. Why does the velocity at the top of a vertical throw vanish while acceleration remains nonzero?

What a full answer mentions: integrate signed velocity; split at roots of v; add absolute path changes; acceleration is still gravity at the turning point.

Week 2: Explain why changing r while $\dot{\theta}$ is constant produces a θ-component of acceleration. When is a_n zero? When does a_r equal $-r\omega^2$?

What a full answer mentions: rotating basis vectors; $2\dot{r}\dot{\theta}$; straight path implies $\rho \to \infty$; constant r gives $\dot{r}=\ddot{r}=0$ and $a_r=-r\omega^2$.

Week 3: Why can normal reaction differ from mg? Why is m a not drawn as an external physical force? For a hill, which direction is positive n?

What a full answer mentions: other vertical force components and curvature; separate kinetic diagram; n toward centre of curvature.

Week 4: Why does tension sometimes contribute no net work in a full ideal pulley system? Why is the energy equation often easier than force equations for final speed?

What a full answer mentions: internal forces cancel through consistent constraints; energy connects initial/final states without solving for time history.

Week 5: Why do you subtract the unstretched length in spring energy? Which energy terms change sign when the datum shifts? Why does v = 0 at maximum spring compression not imply a = 0?

What a full answer mentions: elastic deformation, not total length; only gravitational potential zero shifts consistently; net restoring force can be nonzero.

Week 6 + angular: Why can linear momentum be conserved in an explosion while kinetic energy increases? How can a central force do work despite having zero moment about the centre?

What a full answer mentions: zero external impulse during explosion, internal stored energy converted; radial force does work during radial displacement but has zero torque about the chosen centre.

> **Last-night study rule:** do one question from each major method *without* the solutions; compare your axes, governing equations and constraints, not just the final numerical value. Then practise locating each formula on your two-sided reference sheet. This is more valuable than re-reading an equation list repeatedly.

Coverage audit and source mapping

The two-page reference sheet covers the formula families appearing in the provided lecture sets and the extra angular-momentum material in Tutorial 6. This table shows what was checked so omissions are visible rather than assumed away.

Source	Concepts included	Core models in guide
Week 1 / Tutorial 1	Dynamics definitions, frames, rectilinear motion, average vs instantaneous, variable/constant a, curvilinear vectors, chain rule.	1–4
Week 2 / Tutorial 2	Cartesian, n–t, curvature, polar, rotating basis, Coriolis term, relative translation, projectiles, rotating systems.	4–7
Week 3 / Tutorial 3	Newton, D'Alembert, FBDs, contact/friction/springs, Cartesian/n–t/polar kinetics, mass-center/system forces.	8–10
Week 4 / Tutorial 4 Q4–5	Work for all force types, variable work, kinetic energy, system work, power and efficiency, pulley constraints.	11–12
Week 5 / Tutorial 5	Conservative force, gravitational/elastic potential, general energy, systems and compression; momentum of fragments.	13, 15
Week 6 / Tutorial 6	Linear momentum, linear impulse, componentwise system conservation, angular momentum/cross products and central-force cases.	14, 16–17

Limit of supplied Tutorial 4: only questions 4 and 5 were uploaded as a solved tutorial; the other Week 4 content is supported by the lecture slides. **Limit of angular theory:** Tutorial 6 contains angular momentum applications, while Week 6 lecture slides explicitly teach linear impulse–momentum. The angular section is drawn from the provided tutorial, not invented as Week 6 lecture material.

Reference note: lecture slides contain a few typographical/formatting slips (such as units on acceleration or truncated superscripts). This guide consistently expresses the underlying mechanical relationships and flags potentially confusing instances rather than reproducing a dimensional error.

Edition note. Expanded from the supplied Week 1–6 slides, Tutorial 1–6 solutions and the earlier reference sheet. Added derivations and general method explanations are explanatory reconstructions of those source methods, not new examinable lecture topics. Where a problem depends on an unreproduced figure, consult the named slide/tutorial page for geometry.